GenAI Impact on Data Management

KPMG Inquiry

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Lynne Schneider
February 2024

Topics to be covered

Enterprise Intelligence Architecture

- MDM & Metadata Management
- Data Observability
- Data Quality
- Cloud providers absorbing Fabric/Mesh/Niche capabilities

Data Products & Marketplaces

Organizational shifts in the market (CDOs, Prompt Engineering, Emergent Roles, etc.)

Q & A

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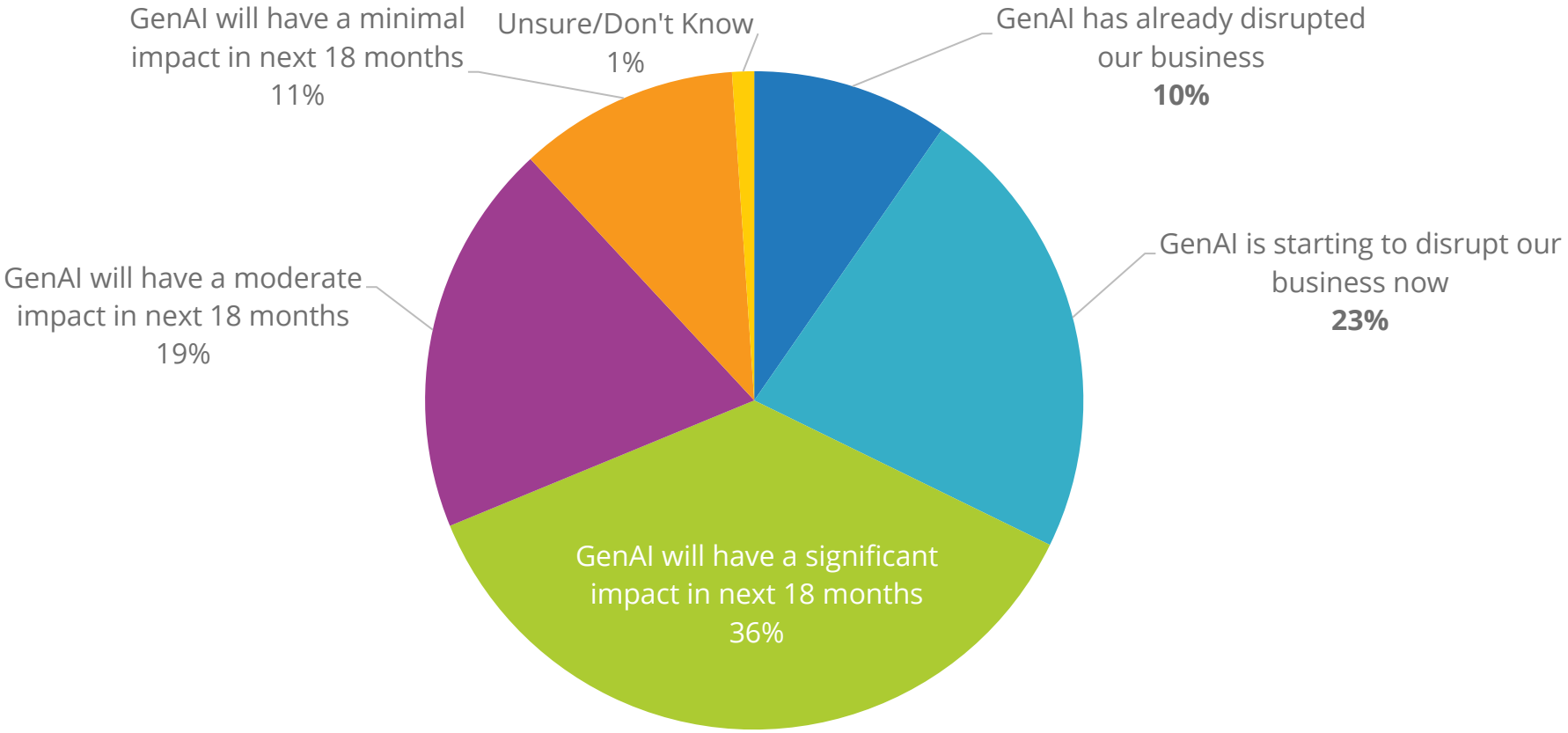
Why are we having this call?

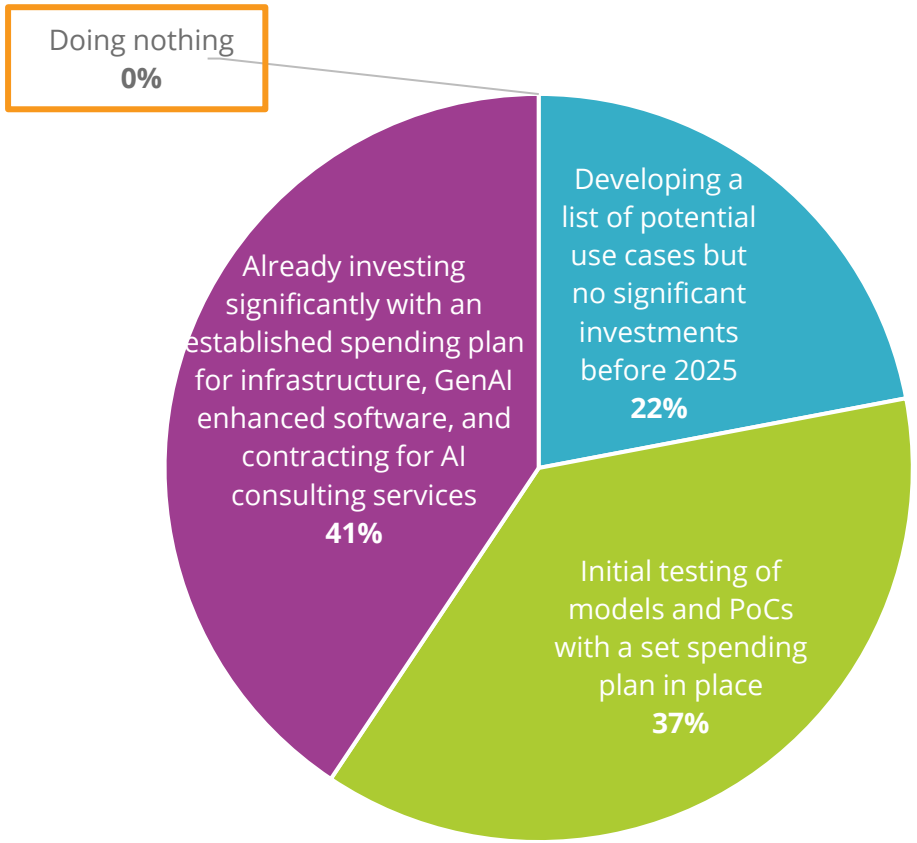
Q. To what extent do you believe broad access to GenAI foundation models, platforms, and application technologies will disrupt your organization's competitive position or business operating model?



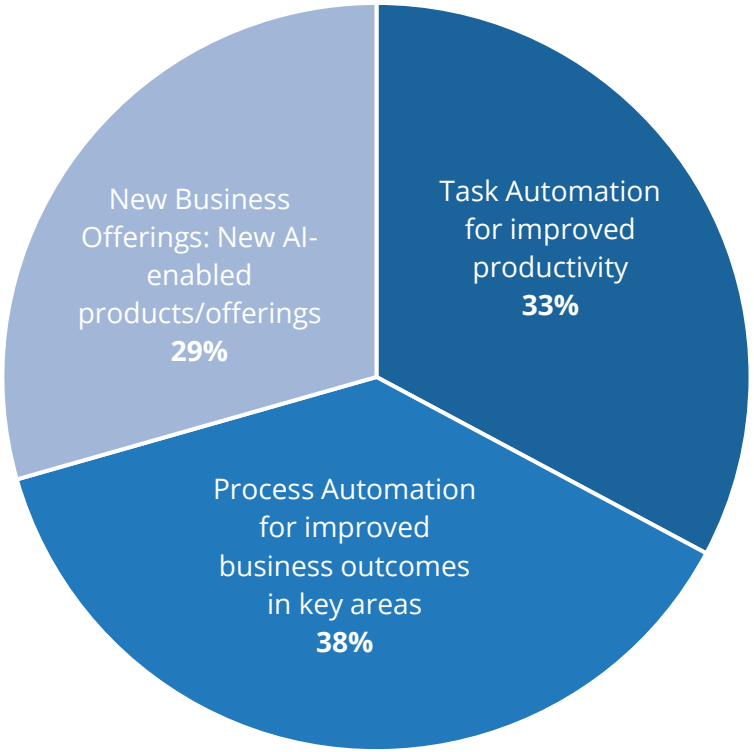
Source: Future Enterprise Resiliency & Spending Survey Wave 1, IDC, January, 2024, N=881

All organizations are doing something with GenAI; Focus is on automation and new opportunities

Q. What is your organization's plan for evaluating or using Generative AI (GenAI) in the next 18 months?



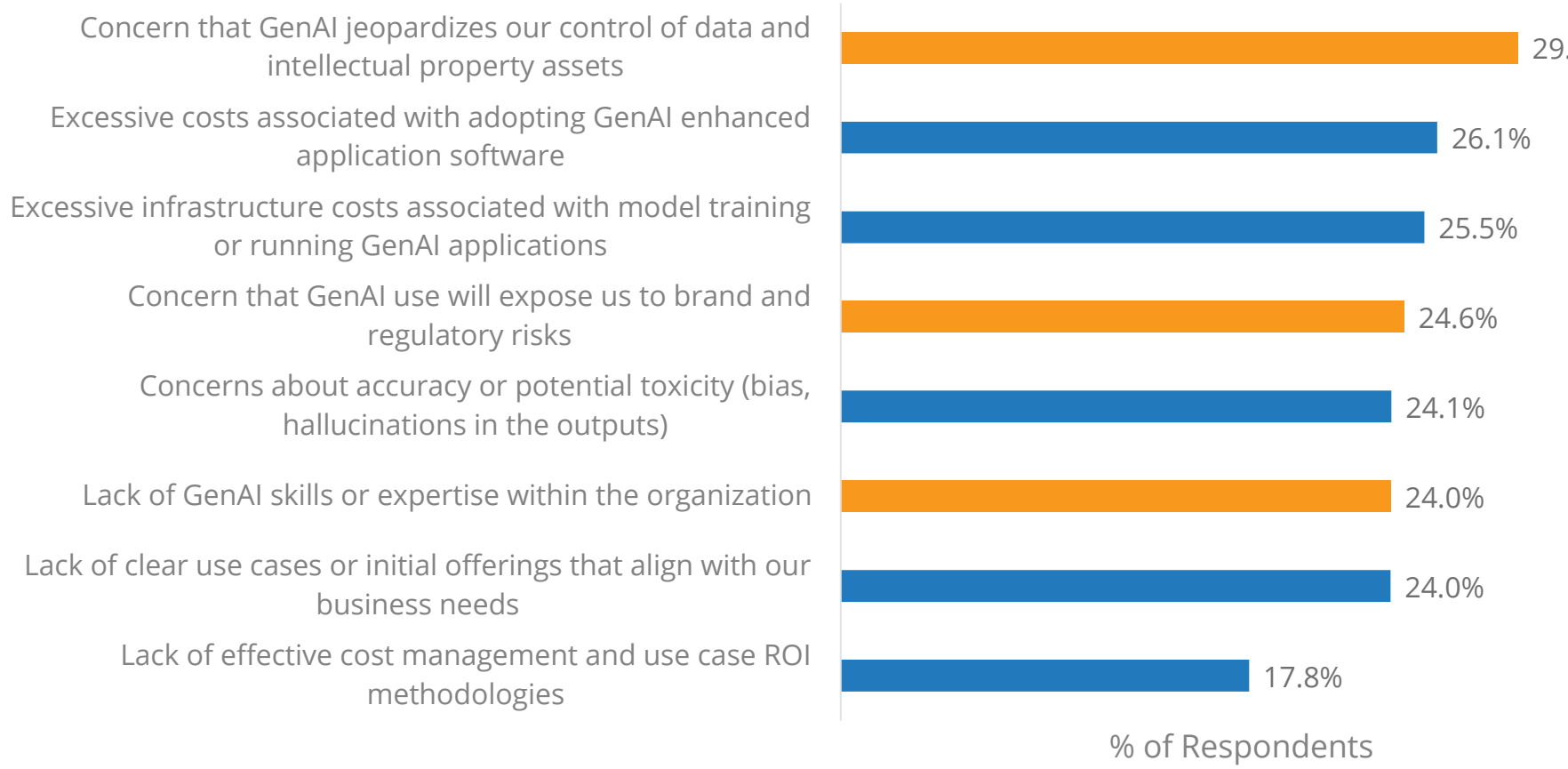
Q. What is your best estimate of how these investments will be allocated in 2024 to the following?



Source: Future Enterprise Resiliency & Spending Survey Wave 1, IDC, January, 2024, N=881

Concerns about data control is the largest constraint being faced; Not far behind is brand and regulatory risks, and skills

Q. What two factors are most significantly limiting further evaluation or expanded use of GenAI in your organization?

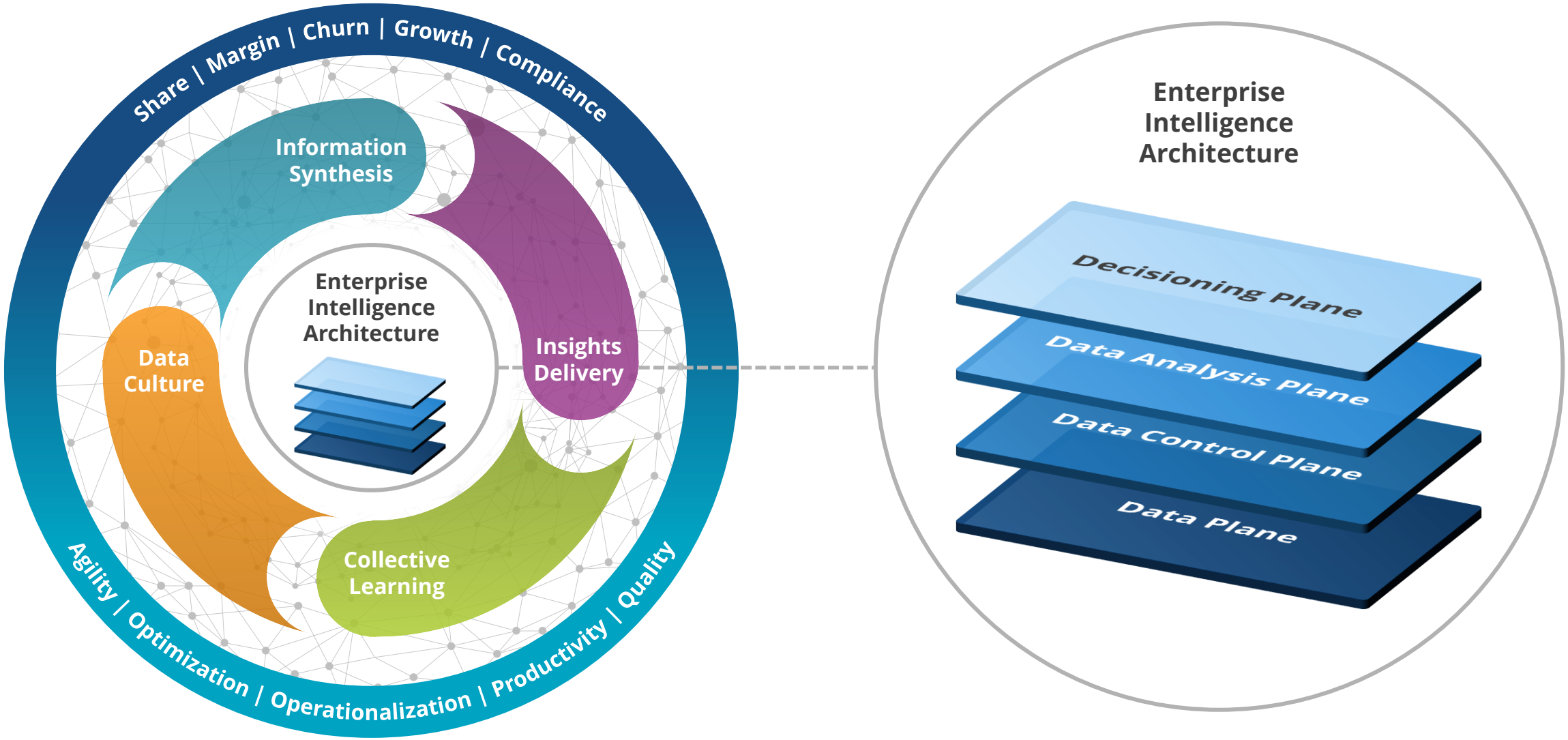


By 2026, 60% of enterprises will squander the GenAI opportunity by

- **Failing to see GenAI artefacts as assets**
- **Failing to weave data, AI and developer perspectives**
- **Failing to embrace new data value chains**

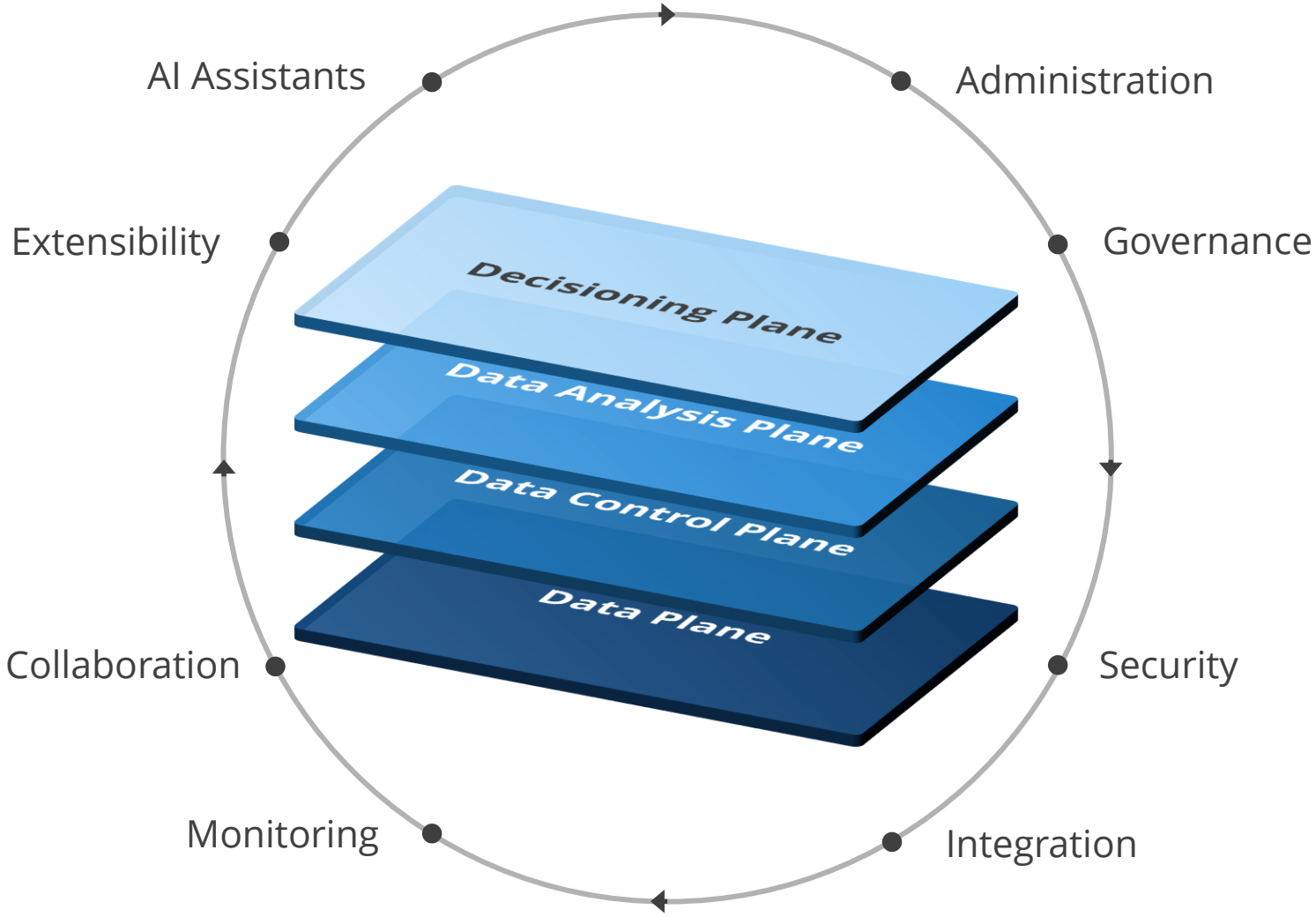
Source: Future Enterprise Resiliency & Spending Survey Wave 1, IDC, January, 2024, N=881

Enterprise Intelligence



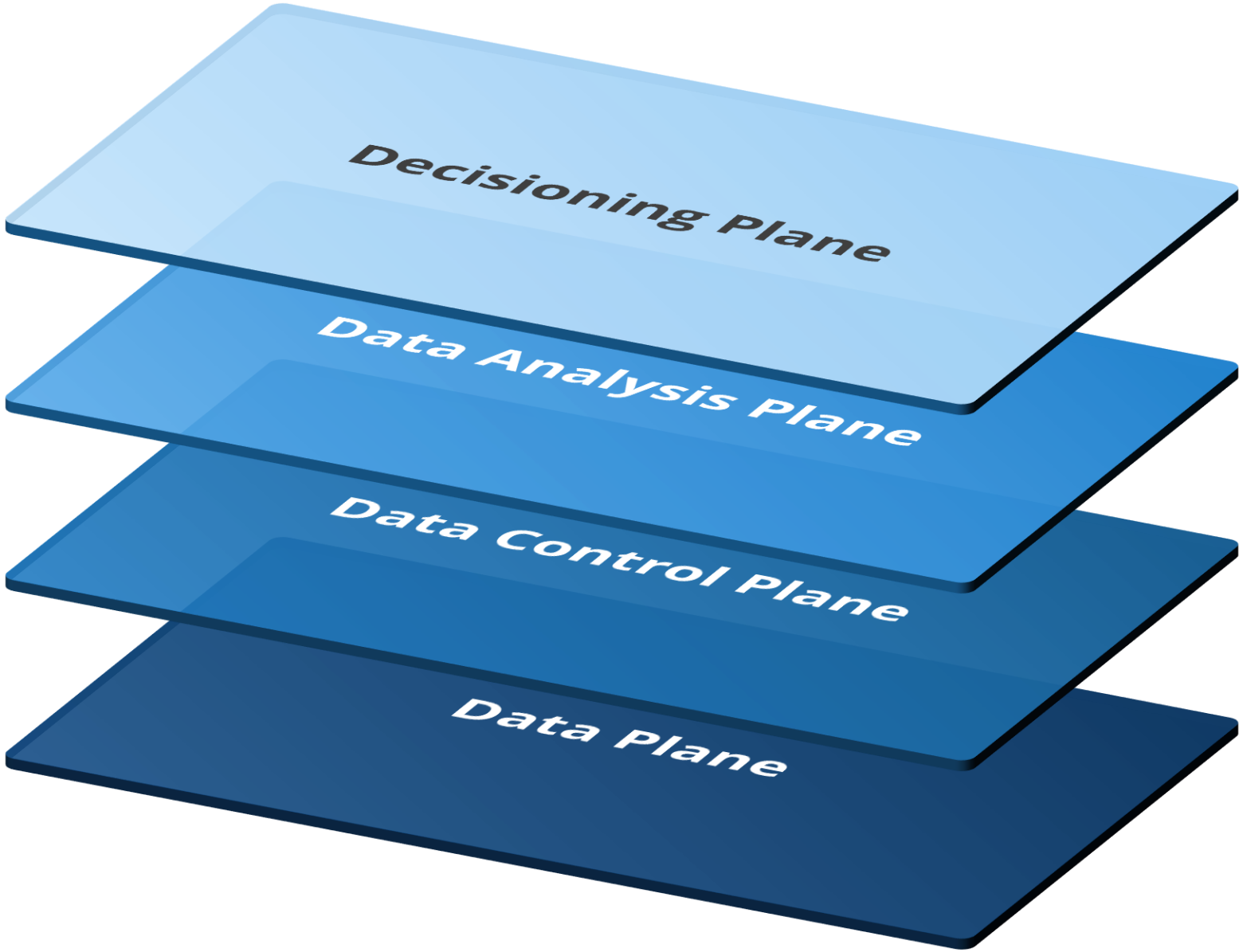
Source: IDC, 2024

Shared Capabilities



Source: IDC, 2024

Enterprise Intelligence Architecture



Source: IDC, 2024

Data Plane

Distributed Data	Domains	Storage	Location	Organization
Diverse Data	Type	Source	Format	Mode
Dynamic Data	Change	Frequency	Event	Life Cycle

Definitions

Source: IDC, 2024

Data Plane
Technology and Functionality

Relational Transaction Processing Database	Document Database	Key Value Store	
Graph Database	Navigational Database	Wide Column Store	Data Lake
Data Warehouse	Data Lakehouse	Streaming Data Processing	Vector Database
Timeseries Database	Unstructured Data or Content Repositories		
Database Administration and Development	Spatial Data Management		

Note: not an exhaustive list of technologies or functionality

Source: IDC, 2024

Data Control Plane

Data Intelligence	Location	Lineage	Semantics
	Context	Classifications	Profiles
Data Governance	Controls	Policies	Observations
Data Engineering	Movement	Replication	Matching
	Transformation	Federation	Cleansing

Source: IDC, 2024

Definitions

Data Control Plane Technology and Functionality

Metadata Management	Data Marketplace	Policy and Control Management
Data Mastering and Semantic Definition		Pipeline Orchestration
Data Observability	Stream Processing	Data Matching and Cleansing
Data Ingestion and Transformation (ETL and ELT)	Data Virtualization	Data Replication Software (Change-Data-Capture)

Note: not an exhaustive list of technologies or functionality

Source: IDC, 2024

Taking control of data and model assets

Why?

- **40%** believe that implementing data sharing and operations practices that ensure data integrity is the **most** important process/policy for assuring success with GenAI.

Are we ready for data use with GenAI?

- **32%** ensure high quality data
- **37%** strictly control sensitive data
- **32%** understand data with IP ownership issues
- **30%** track and control GenAI use of internal data

How?

- **Data** intelligence connected with **model** intelligence will enable security, privacy and trust.

Source: Future Enterprise Resiliency & Spending Survey Wave 1, IDC, January, 2024, N=881

What (Gen)AI capabilities are emerging in DII software?

Integration

Pipeline Documentation

CoPilots

- Generating one operation in a pipeline
- Generating a pipeline

Prompt engineering

- Turning unstructured into structured
- RAG and vector embedding

Mastering

- Entity and relationship discovery

Intelligence

Data documentation

Lineage inference

Data quality and observability:

- Natural language rule creation
- Anomaly detection

Governance:

- Sensitive data discovery and classification
- Policy creation

Software vendors building data control plane portfolios

Cloud

AWS
- Glue
- Data Zone

MS Azure
- Data Fabric
- One Lake

GCP
- Dataplex
- Dataflow
- Data Fusion

ISVs

Databricks

Informatica

Precisely

Qlik and Talend

SAP (DataSphere)

IBM (CloudPak for Data)

Oracle OCI

IDC Market Glance: Data Control Plane

Intelligence

Metadata Management and Sharing

Acryl Data, AfterDataai, Alation, ataccama, atlan, aws, BigID, celonis, CLOUDERA, Collibra, CONFLUENT, databricks, data.world, dbt, denodo, experian, genpact, Global IDs, Google, HITACHI Inspire the Next, IBM, Informatica, InterSystems, Microsoft, Nexla, OCTOPAI, OneTrust, ORACLE, Orion, OVALEDGE, Palantir, precisely, Promethium, Qlik, Quest, RELTIO, Rocket, SAP, sas, securiti, SELECT STAR, silwood, snowflake, Solidatus, Syniti, tableau, talend, Tamr, teradata, tresata, Modern, THINKDATA WORKS, TIBCO, XTREMEDATA

Mastering and Semantic Definition

AllegroGraph, ataccama, ATSCALE, boomi, CAMBRIDGE SEMANTICS, cinchy, dun&bradstreet, HITACHI Inspire the Next, IBM, Informatica, ORACLE, Profisee, Progress, quantexa, RELTIO, SAP, sas, Semarchy, Senzing, STARDOG, STIBO SYSTEMS, Syndigo, Syniti, Tamr, TIBCO

Governance

Policy and Control

1touch.io, aiimi, Alation, ALTR, ataccama, atlan, aws, ANONOS, BigID, Collibra, databricks, Delphix, IBM, IMMUTA, imperva, Informatica, mage, Microsoft, OneTrust, ORACLE, OVALEDGE, Palantir, PKWARE, precisely, privacera, Promethium, PROTEGRITY, Quest, RELTIO, SAP, satori, securiti, secuvy.ai, skyflow, Solidatus, talend, Modern, THINKDATA WORKS, VARONIS, Velotix

Observation

acceldata, Anomalo, ataccama, Bigeye, Collibra, Datafold, DataKitchen, IBM, Informatica, great expectations, KENSU, lightup, MONTE CARLO, Nexla, precisely, Promethium, sifflet, ScoreData, SODA, Modern, XTREMEDATA

Engineering

Ingestion and Transformation and Virtualization

ACTIAN, Airbyte, ALTAIR, alteryx, aws, Astera, ataccama, ATSCALE, boomi, CHAOSSEARCH, Cleo, CLOUDERA, coalesce, composable, CONFLUENT, databricks, KYLIGENCE, dbt, denodo, dremio, Fivetran, GIGASPACES, Google, GridGain, HASURA, HAZELCAST, Hevo, HITACHI Inspire the Next, Hewlett Packard Enterprise, IBM, INCORTA, Informatica, Infoworks, integrate.io, InterSystems, MATILLION, opentext, Microsoft, MuleSoft, Nexla, ORACLE, Palantir, precisely, Prophecy, Progress, Promethium, Qlik, Red Hat, Rocket, SAP, sas, Semarchy, snaplogic, snowflake, splunk>, Starburst, StreamSets, striim, SYNATIC, Syniti, talend, teradata, Modern, THINKDATA WORKS, TIBCO

Replication

databricks, debezium, Fivetran, Google, IBM, Informatica, integrate.io, MATILLION, Microsoft, ORACLE, precisely, Qlik, Quest, striim, talend

Matching and Cleansing

Anomalo, ataccama, Collibra, DATACTICS, Datafold, dun&bradstreet, experian, genpact, Global IDs, IBM, Informatica, Nexla, OVALEDGE, Palantir, precisely, RELTIO, SAP, sas, Semarchy, Syniti, talend, Tamr, Modern, TIBCO, zoominfo

Orchestration

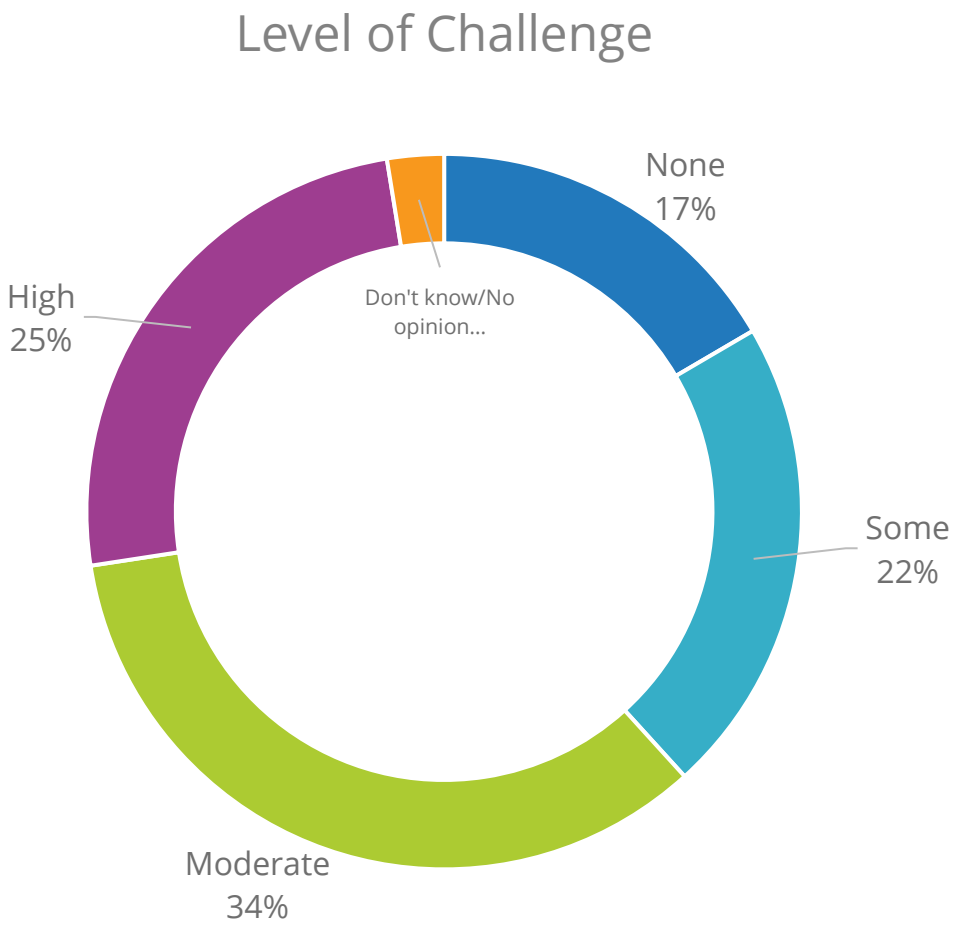
Airflow, Astera, ASTRONOMER, ataccama, ATSCALE, bmc, BROADCOM, composable, dagster, DataKitchen, dbt, Delphix, elementl, Google, IBM, Informatica, Quest, Palantir, Prophecy, Promethium, SOLIX, Syniti, Modern, WhereScape

Source: IDC, 1Q24

For areas on which IDC publishes market share data, the top 3–5 market share leaders are represented. For areas on which IDC does not publish market share data, vendor selection is up to analyst discretion.

One of the biggest issues, even within data marketplaces, has been the search for the 'right' dataset/data product

More than half have a moderate challenge



When sourcing data externally, the most common terms have been data-as-a-service and datasets

Evolving practices with the refinement of internal data sharing have elevated the discussion of 'data products'

In external data marketplaces, the ability to source models and apps has risen sharply over the past 12 months – and all are sometimes covered by the term 'data products'

Source: Data as a Service and Data Marketplaces (Worldwide/North America External Data Users), IDC, April, 2023
N=507
Note: Managed by IDC's Global Primary Research Group.

Data “Marketplace” Market Suffers from a Lack of Crisp Definitions

And we’re not going to solve that here . . .

Leading enterprises share data internally, as well as with their suppliers, customers, financial institutions, regulatory bodies and others – and many have a desire to create data-as-a-service offers for monetization

For all of the data exchanged, it needs to be permissioned in some way. This activity should be tracked and governed, whether the second party is accessing the data in a shared location, via API or other solution, or even a file sent to that party

Additional functionality is needed if the usage is going to be metered. This functionality can be useful if for tracking and is necessary if there is going to be access or usage volume billing.

New use cases are coming to the forefront that combine the sharing, exchanging and collaboration with data that require increasingly strict levels of data privacy

Framework for Discussing a Hierarchy of Data Exchange and Collaboration

Some platforms offer multiple levels of functionality

How is Generative AI Changing Data Marketplaces?

All are improvement in the speed to new insights

Minimizes the need for creating ontologies and thematic catalogs

Changes the nature of search to make discovery of data more accessible via natural language

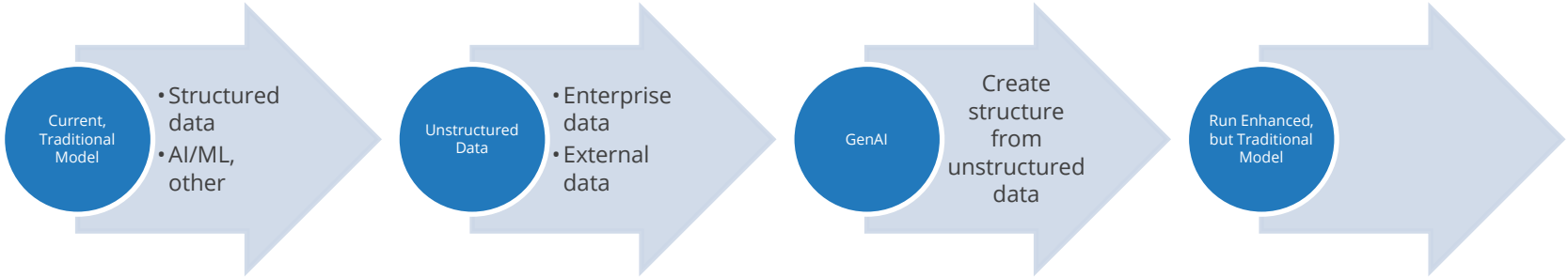
Assists with 'leveling up' usage by non-data scientists

Matches of data and algorithms

S&P Global Launches Generative AI Search on the S&P Global Marketplace

Use of Generative AI Will Lead to More Datasets, More Rapidly

Improve analyses already common today



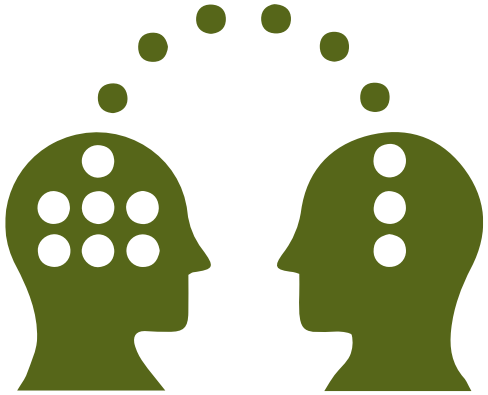
GenAI is also helping to create new datasets from unstructured data that can be turned into structured datasets for more traditional analyses.

For example, it is difficult to combine a dataset of customer transactions with customer services notes/ transcripts.

With GenAI, this can be used to curate the unstructured data so that it can be added to a model of customer churn, for example.

The same principle applies to creating data products and creating meta data and other information from structured and unstructured data internally and for external monetization

Investment In People Is a Critical Success Factor



30%

Of new GenAI investment is in people

36%

Creating a GenAI center of excellence

31%

Creating mandatory GenAI awareness and acceptable use training

Source: GenAI ARC Survey, IDC, August 2023, N=1,363 Data weighted by IT spend

24%

Organizations that believe their resources are extremely prepared for GenAI

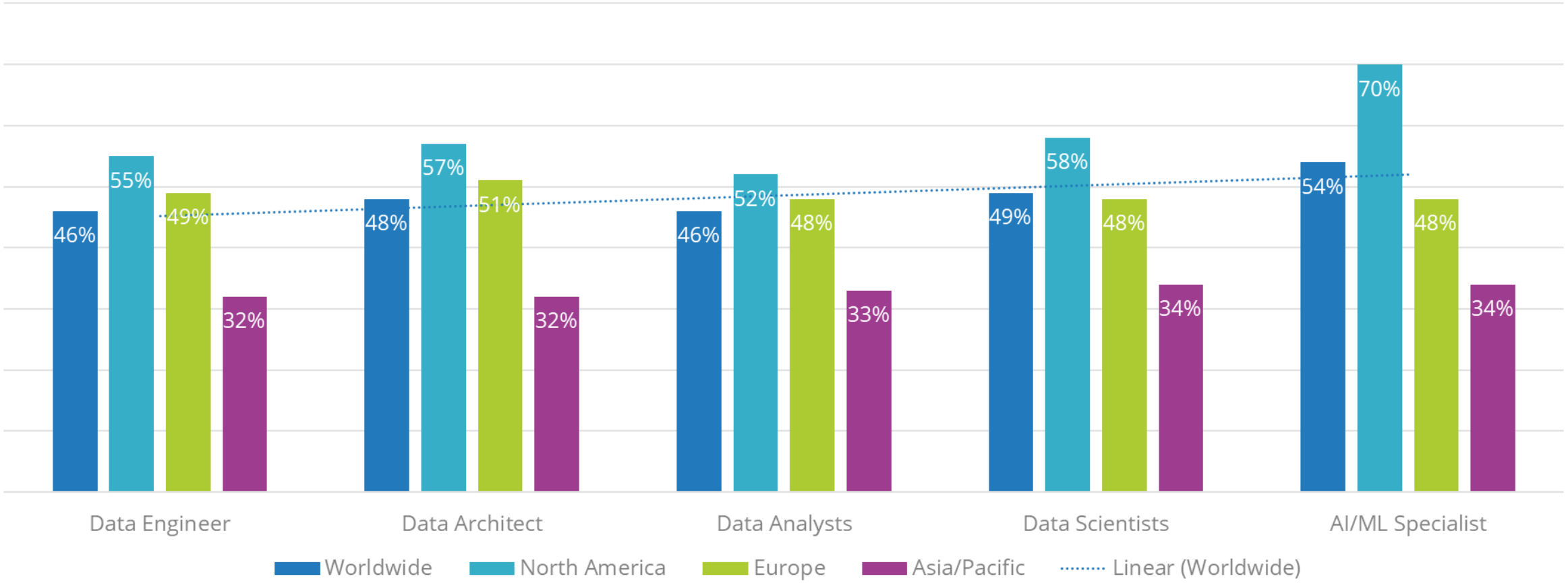
Source: Future Enterprise Resiliency & Spending Survey Wave 1, IDC, January, 2024, N=881

IDC Survey Spotlight

What AI and Data Roles Are Most Difficult to Fill at Enterprises Worldwide?

Gina Smith

Q. How would you rate your organization's ability to hire for the following roles? (DIFFICULT and VERY DIFFICULT)



IDC #US50846023 (June 2023)
Source: IDC Future Enterprise Spending Survey, Wave 1. Feb 2023. N = 985 worldwide (North America N = 354, Europe N = 225, Asia/Pacific N = 300

Shifts in the market

“By 2027, GenAI will help equalize spending on unstructured and structured data processing and analysis software, doubling unstructured data's productive use.”

“By 2028, the tenure of the average CDO will at least double, reflecting business executives evolving understanding of the path to greater enterprise intelligence, data, and AI value creation.”

“To mitigate new risks created by the pervasive use of generative AI by their workforces, by the end of 2025, 60% of large enterprises will mandate formal data literacy and responsible AI training.”

“By 2027, 40% of current job roles will be redefined or eliminated across G2000 organizations accelerated by GenAI adoption.”

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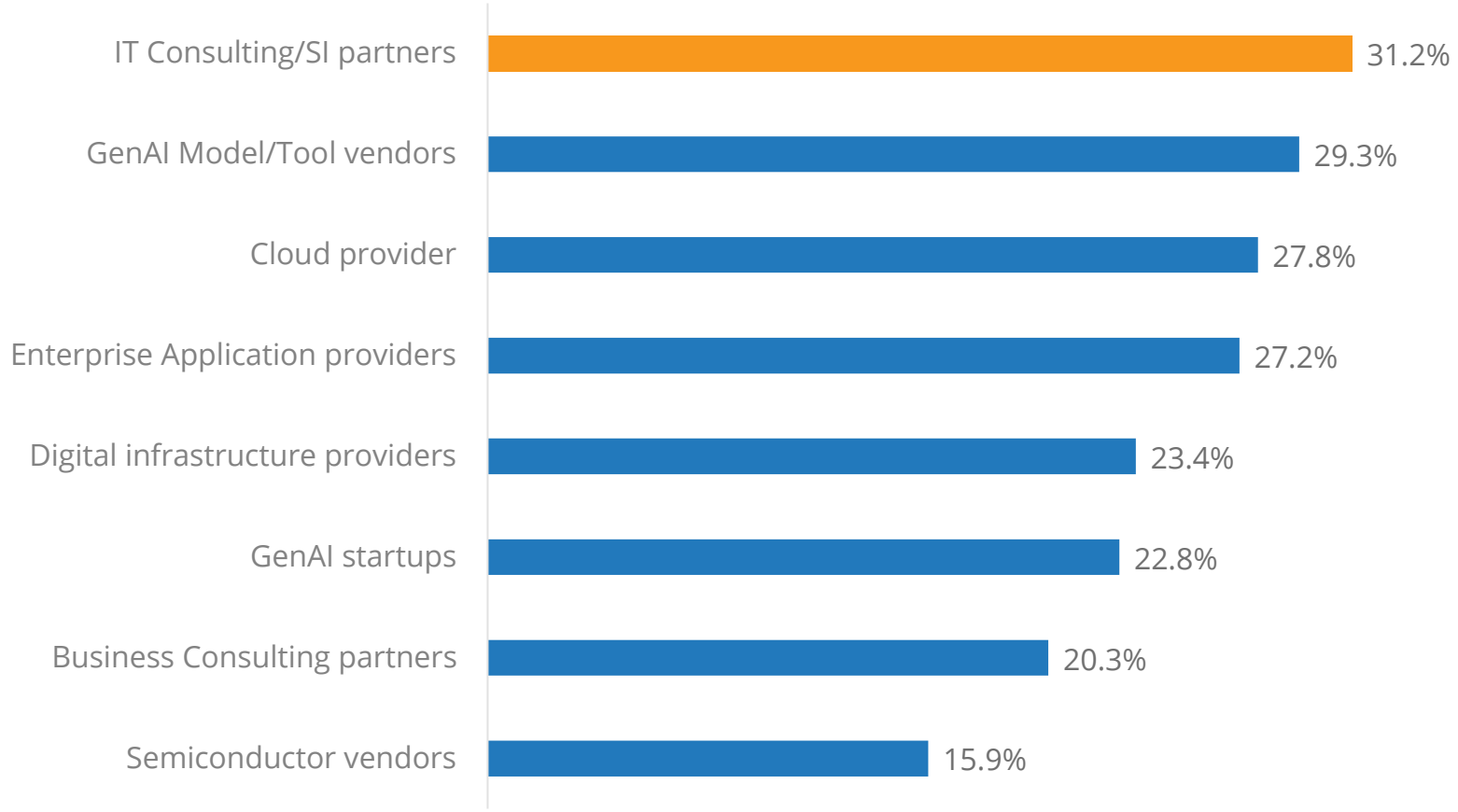
IT consulting / SI partners are the most important strategic GenAI technology provider this year

Q. Which of the following type of vendor will be your most important strategic GenAI technology partner in the next 12 months?



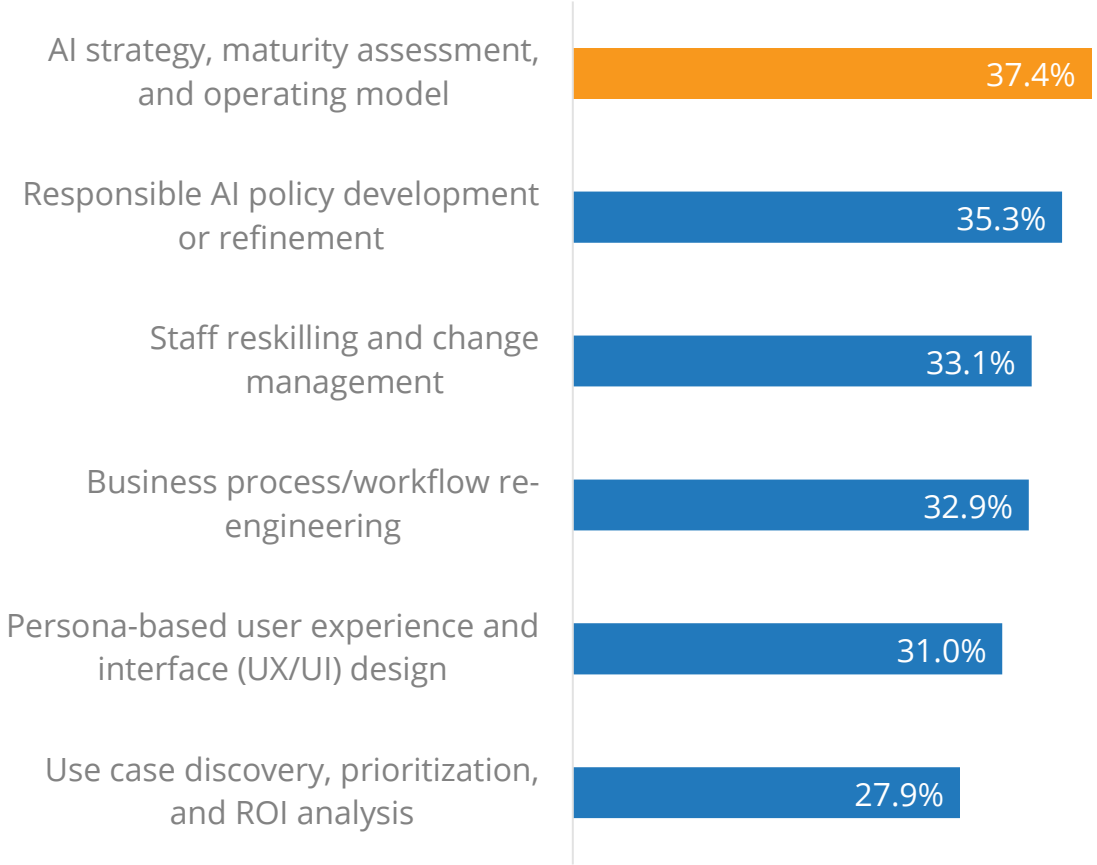
51% perceive external AI enabled services as value-add, delivered faster, at greater volume, and with better quality and insight than with human-based delivery alone.

34% of organizations expect to increase spending on external services as AI brings new issues to address

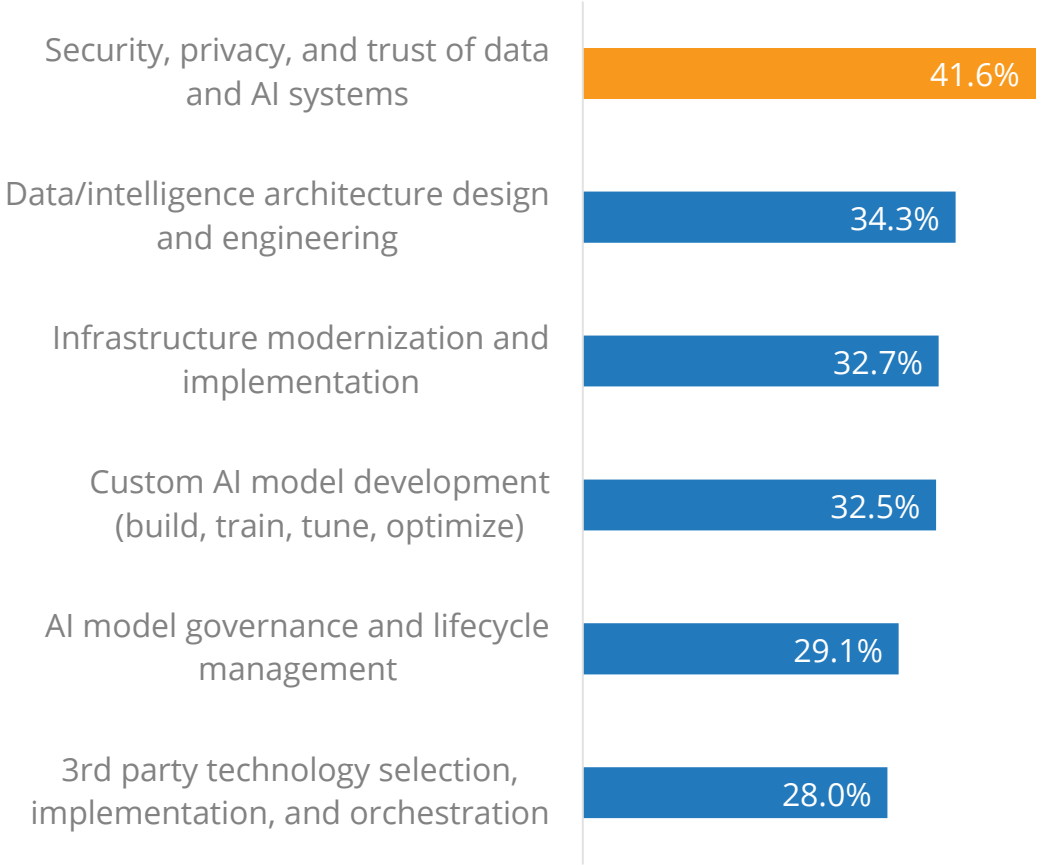
Source: Future Enterprise Resiliency & Spending Survey Wave 1, IDC, January, 2024, N=881

Strategy and data top the list of focus areas that organizations are looking to service providers for help

Q. Which AI initiative business-focus areas?



Q. Which AI initiative technical-focus areas?



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